

Notes

Google Class Room

Name of the Module : JSE (Java Standard Edition)

Duration /Session : 101 Session (4 Months) (60% Core Java -> ADV Java)

Sunday classes : Online Sunday classes (9 -10 sessions) [Helping Topics OR Common topic]

Java version : Java Java 17

Lab assignment : 1 and provide all the Assignments (MCQs + Sample coding + Scenario based)

Validity Of Test : 1 day [11AM]

Lab is compulsory : Without Lab, Permission is not for Core Java classes

After course : 10-year developer knowledge

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What is a language ?

* A language is a **communication media** through which we can communicate with each other.

What is a Programming Language ?



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graph LR; User((User)) -- "User wants to communicate in terms of Java Language" --> JavaLanguage[Java Language]; JavaLanguage --> Computer((Computer))
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* A Programming language is an intermediate between the user and computer machine. By using programming language we can instruct the instructions to get the desired output.

Characteristics of the programming language :

* We have 2 characteristics :

1) Syntax (Rules given by that particular language)

2) Semantics (Meaning OR Structure given by that particular language)

Example :

English Translation :

Syntax [Subject + Verb + Object]

We like a boy. (Valid)

We like fish. (Invalid)

* In our Java programming Language we have 2 levels of security :

1) At compiler level :

When our Java compiler will only **verify** whether the programmer is following the syntax or not ? Compiler will check rules and ignoring the execution of the program. The compiler will generate syntax error if something is wrong in the code.

2) At runtime level :

Now the runtime environment will verify whether the code is meaningful OR meaningful. If the code is meaningful then it will generate a runtime error.

So, the final conclusion is : Java compiler will verify the syntax and our runtime environment will verify the semantics (meaning OR structure of the code)

What is Java ?

* Java is a high-level, platform-independent, secure, robust (strong), multithreaded and object-oriented programming language.

* Java was originally developed by Sun Microsystems in 1995 for the first version of Java (or JDK) (Java Development Kit), 1.0 was released on 23rd January 1996.

* In the year 2010, Java was acquired by Oracle Corporation.

* By using Java language, we can develop various applications like :

1) Stand-alone application (Desktop OR Software)

2) Web application (Servlets)

3) Enterprise applications (JSP/JSP/JSP)

4) Mobile applications

5) Distributed applications